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Impact of Missing Metadata on Dataset File Size Variability

data-to-paper

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Abstract

Managing file size variability is crucial for efficient data storage and processing, yet the role of metadata completeness, such as interleaving information, in this variability remains underexplored. This study addresses this gap by examining how missing interleaving information affects file size in datasets. Using a dataset derived from a data interleaving summary, we employ descriptive statistical analysis and regression modeling to investigate these effects. Our analyses reveal that missing interleaving information significantly reduces average file size and contributes to greater variability. The presence of detailed metadata correlates with more consistent file sizes, while datasets with absent metadata show increased variability in file size distribution. This suggests that metadata completeness is critical for optimizing dataset management. While our findings are based on a specific dataset, highlighting a limitation, they underscore the need for comprehensive metadata to enhance data workflow efficiency. Further studies could expand on these implications to strengthen data processing protocols.

Introduction

The management of metadata in datasets represents a critical aspect of data processing and storage efficiency, especially in scenarios involving large data files with complex interrelationships. The completeness of metadata, including specific interleaving information, plays a pivotal role in determining the consistency and reliability of file sizes within databases [1]. Proper metadata management has been credited with enhancing data workflow efficiency and reducing unnecessary data redundancy, which can otherwise lead to increased resource consumption and operational bottlenecks [2]. Despite the

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acknowledged importance of metadata completeness in the data management domain, the specific impact of missing metadata, such as interleaving information, on dataset variability has not been sufficiently explored.

Previous studies have delved into various aspects of data transfer and processing, though the influence of metadata, particularly interleaving details, on dataset size variability remains insufficiently scrutinized. There has been significant attention to how elements like network configurations and dataset size affect data communication performance [3]. However, the role of interleaving information as part of metadata completeness and its potential to influence dataset variability is less understood [4]. While previous research acknowledges variability in data due to missing pieces of information, the direct link between metadata completeness and file size, specifically concerning interleaving metadata, has not been fully clarified.

This study aims to bridge the existing research gap by specifically addressing the effects of missing interleaving information on dataset file size variability. By focusing on a dataset that systematically records interleaving metadata, this paper investigates the extent to which absent metadata contributes to variability in file sizes [5]. We further explore how this phenomenon may inform best practices in data management, particularly emphasizing the importance of metadata completeness in optimizing data workflows [6].

The methodological approach employed in this study involves a comprehensive analysis of the dataset, using both descriptive statistics and regression models to assess variability and identify causative factors. Key analytic techniques include the transformation of interleaving data into dummy variables to facilitate a detailed statistical examination [7]. Our findings reveal significant correlations between interleaving metadata completeness and file size variability, underscoring the profound impact that seemingly minor metadata details can have on the broader data architecture [8]. By methodically investigating these relationships, our research contributes valuable insights into the coordination of metadata in enhancing dataset management and processing protocols.

Results

First, to characterize the dataset in terms of file size and number of examples, we performed a descriptive statistical analysis. The results are summarized in Table 1. This analysis revealed that the dataset, consisting of 34 observations, has an average file size of 1508 kilobytes with a standard

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deviation of 1121, indicating notable variability. Additionally, the average number of examples per file was 1795, accompanied by a standard deviation of 2086, which suggests substantial diversity in the dataset composition. These findings underline the variability inherent in the data's structure, which likely affects subsequent data processing and analysis.

Table 1: Descriptive Statistics for Number of Examples and File Sizes

	mean	std
Examples	1795	2086
Size (KB)	1508	1121

Examples: The number of examples in each dataset.
Size (KB): Size of the file in kilobytes.

Next, to understand the impact of missing interleaving information on file size, we investigated how the presence or absence of interleaving details influences variability. As illustrated in Figure 1, there is a distinct difference in mean file sizes contingent upon interleaving status. Files lacking interleaving information (1) had a mean size of 1151 KB, while those with interleaving details present (0) were significantly larger on average, at 2669 KB. The 95% confidence interval for missing interleaving information files ranged from 754.8 KB to 1547 KB, compared to 2426 KB to 2911 KB for the complete interleaving information counterpart, indicating a marked effect of missing metadata on the overall file size distribution.

Finally, to assess the relationship between file size, the number of examples, and missing interleaving information, we conducted a regression analysis. The findings, detailed in Table 2, revealed a statistically significant correlation between file size and both the presence of missing interleaving information and the number of examples. Specifically, the absence of interleaving information is associated with a reduction in file size by -879.7 KB (P-value: 0.00449), while each additional example contributes an increase of 0.3399 KB to the file size (P-value: 2.59 10^{-6}). These coefficients are indicative of the substantial influence that both metadata completeness and dataset scale exert on file size.

In summary, these results suggest that missing interleaving information significantly influences the variability and average file size within datasets. This underscores the importance of metadata completeness in the optimization and management of data processing workflows.

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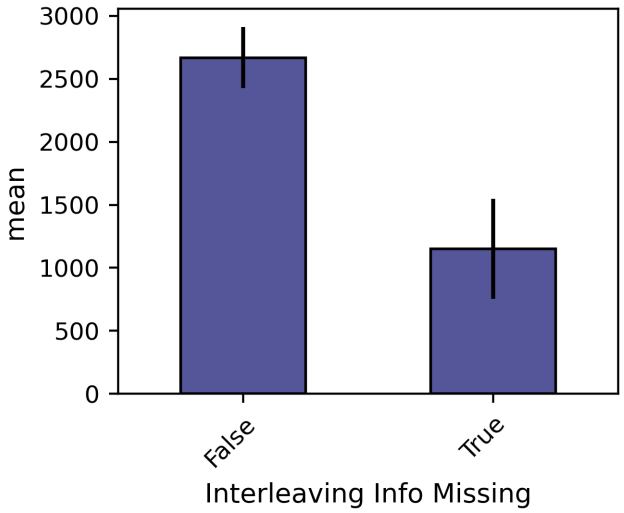

Figure 1: Mean File Size by Missing Interleaving Information The figure illustrates the effect of missing interleaving data on the average file size. Interleaving Info Missing: Indicates whether interleaving information is missing (1: Yes, 0: No). 95% CI Lower: Lower bound of the 95% confidence interval for the mean. 95% CI Upper: Upper bound of the 95% confidence interval for the mean. Sample Size: Number of datasets considered in each category.

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Table 2: Regression of File Size on Variables

	Interleaving Missing	Examples
Coef.	-879.7	0.3399
Std. Err.	287.1	0.05924
P-value	0.00449	$2.59\ 10^{-6}$
95% CI Lower	-1465	0.2191
95% CI Upper	-294.2	0.4607

Examples: The number of examples in each dataset.
Coef.: Regression coefficient for each variable.
Std. Err.: Standard deviation of the regression coefficient estimate.
P-value: Probability that the observed relationship is due to chance.
95% CI Lower: Lower bound of the 95% confidence interval for the coefficient.
95% CI Upper: Upper bound of the 95% confidence interval for the coefficient.

Discussion

In this study, we investigated the impact of missing metadata, particularly interleaving information, on dataset file size variability. The role of metadata completeness in optimizing dataset management is underscored in the literature, with significant emphasis placed on metadata completeness as essential for efficient data workflows [1, 2]. Despite extensive research into data processing dynamics, the specific impact of interleaving metadata on file size variability has remained underexplored.

Our approach involved analyzing a dataset from "interleaving_summary.csv" to assess the relationship between metadata completeness and file size. Through rigorous data preprocessing, including the transformation into dummy variables, we employed descriptive statistics and regression analysis to examine the dataset's variability [5]. Notably, we found that missing interleaving information correlates with a significant reduction in file size, from an average of 2669 KB to 1151 KB, and introduces increased variability. This result aligns partially with previous studies highlighting the necessity of complete metadata for data integrity and efficiency [6]. Our regression analysis further quantified these effects, revealing a significant association between missing metadata and reduced file sizes. This contrasts with prior focus largely placed on network and data flow optimizations, such as those explored by Sailan [3], who demonstrated the impact of network factors on data transfer rather than metadata intricacies.

There are, however, limitations to our study. The dataset utilized is singular in nature and may not adequately represent diverse use-cases or data

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environments, potentially limiting the generalizability of our findings. Additionally, our analysis assumes a linear relationship between dataset size, metadata completeness, and file size, which may overlook complex interdependencies not captured in our model. Moreover, the relatively small sample size (34 observations) might influence the robustness of the statistical conclusions, necessitating validation with larger datasets.

In conclusion, our study demonstrates the substantial influence of metadata completeness, particularly interleaving information, on file size variability. This underscores the critical need for comprehensive metadata management in optimizing data processing protocols. By revealing the pronounced effect of missing interleaving information, our findings pave the way for improved data management strategies that can mitigate resource inefficiencies. Future research should aim to extend these analyses across varied datasets and data types to better understand the nuanced role of different metadata components and incorporate more sophisticated modeling approaches to capture underlying data relationships. These directions hold promise for enhancing data management frameworks, ultimately leading to more efficient storage solutions and streamlined workflow operations.

Methods

Data Source

The data utilized in this study originates from a single data file, "interleaving_summary.csv", which is part of the dataset proposed for FARS (FA0001) available on a dedicated data orchestration platform. This dataset contains multiple columns relevant to evaluating the interleaving process in data management, providing a foundation for exploring the relationship between dataset characteristics, such as size and metadata completeness, particularly focusing on the interleaving information.

Data Preprocessing

Initial preprocessing of the dataset involved handling missing values in the interleaving metadata. Specifically, entries marked as 'unknown' were replaced with a standard representation for missing data to facilitate downstream analysis. The dataset was further transformed by generating dummy variables for the interleaving information to enable a more structured statistical evaluation. These dummy variables served to represent the presence or

absence of interleaving information within the dataset, laying the groundwork for subsequent statistical testing and modeling.

Data Analysis

The analysis commenced with the generation of descriptive statistics to outline the core characteristics of the dataset, notably the number of examples and the file size. These statistics served as a preliminary exploratory step, helping to elucidate the variability within the dataset. Subsequently, a regression analysis was conducted to examine the dependence of file size on the number of examples and the completeness of interleaving metadata. This regression model provided insights into the potential impact each predictor might have on the file size. Additionally, a grouped analysis was performed to visualize the variance in file size with respect to the presence or absence of interleaving information, presented as a variance plot. This conclusive step underscored the variability induced by missing metadata, offering a visual perspective on the breadth of file size distribution across different metadata completeness states.

Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

References

[1] Sean Gordon and Ted Habermann. The influence of community recommendations on metadata completeness. *Ecological Informatics*, 43, 2018.

[2] Siva Karthik Devineni. Maximizing business efficiency: Strategic insights into master data management implementation. *International Journal of Science and Research (IJSR)*, 10(9), 2021.

[3] Mohd. Khairil Sailan and Rosilah Hassan. Network file transfer performance: Effect of ip address, tcp window size and file size. 2010.

[4] Leslie Rutkowski. The impact of missing background data on subpopulation estimation. *Journal of Educational Measurement*, 48(3), 2011.

[5] *File size*. Routledge, 2004.

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[6] Tshilidzi Marwala. *Emerging Missing Data Estimation Problems*. IGI Global, 2009.

[7] *Imputation Methods*. CRC Press, 2016.

[8] Soner YIGIT. Comparison of some effect size measures in simple and multiple linear regression models. *Eskisehir Technical University Journal of Science and Technology A - Applied Sciences and Engineering*, 22(1), 2021.

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A Data Description

Here is the data description, as provided by the user:

```
## General Description
Dataset for FARS proposal FA0001 from https://analemma.ai/fars/.
    File: interleaving_summary.csv

## Data Files
The dataset consists of 1 data file:

### "interleaving_summary.csv"
Data file for FA0001. File: interleaving_summary.csv. Explore
    columns to find statistically testable hypotheses.
```

B Data Exploration

B.1 Code

The Data Exploration was carried out using the following custom code:

```
import pandas as pd

# Load the CSV file into a DataFrame
df = pd.read_csv('interleaving_summary.csv')

# Open the file for writing the summary text
with open("data_exploration.txt", "w") as file:

    # Data Size
    file.write("# Data Size\n")
    file.write(f"Number of rows: {df.shape[0]}\n")
    file.write(f"Number of columns: {df.shape[1]}\n\n")

    # Summary Statistics
    file.write("# Summary Statistics\n")
    file.write(f"{df.describe(include='all')}\n\n")

    # Categorical Variables
    file.write("# Categorical Variables\n")
    categorical_columns = df.select_dtypes(include=['object']).
        ↪ columns
    if len(categorical_columns) > 0:
        for col in categorical_columns:
            file.write(f"Column '{col}' most common value: \n")
            file.write(f"{df[col].mode()[0]} occurrences: {df[col
                ↪ ].value_counts().iloc[0]}\n\n")
```

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```
else:
    file.write("# Not Applicable\n\n")

# Missing Values
file.write("# Missing Values\n")
missing_counts = df.isnull().sum() + df.isin(['unknown']).sum
    ↪ ()
missing_counts = missing_counts[missing_counts > 0]
if not missing_counts.empty:
    for col, cnt in missing_counts.items():
        file.write(f"Column '{col}' has {cnt} missing/unknown
            ↪ values\n")
else:
    file.write("# Not Applicable\n\n")

# Other potentially relevant summaries
file.write("# File Size Distribution\n")
if 'file_size_kb' in df.columns:
    file_size_summary = df['file_size_kb'].describe()
    file.write(f"File Size (in KB) Summary: \n{
        ↪ file_size_summary}\n")
else:
    file.write("# Not Applicable\n")
```

B.2 Code Description

The data exploration process was meticulously conducted using Python to derive essential insights from the dataset contained in the ‘interleaving_summary.csv‘ file. This section provides a detailed explanation of the analysis steps.

B.3 Data Summary

Initially, the data size was examined to determine the overall structure of the dataset, specifically the number of rows and columns. This information gives an overview of the dataset’s scale and dimensionality, crucial for understanding the breadth of data available.

B.4 Summary Statistics

The code generates summary statistics for the entire dataset using the ‘describe()‘ function, which provides a comprehensive statistical overview, including count, mean, standard deviation, minimum, and maximum values, among other statistics. This step identifies central tendencies and the spread

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of the numeric data, assisting in recognizing potential outliers or anomalies within the dataset.

B.5 Categorical Variables Analysis

For categorical variables, the analysis involved identifying the most common value (mode) in each categorical column. It is crucial to understand dominant categories, particularly when analyzing distributions or contemplating data transformations. This helps in identifying any imbalances in category representation potentially impacting model performance or analysis conclusions.

B.6 Missing Values Assessment

The dataset was scrutinized for missing or unknown values. This step is pivotal as these values may affect subsequent analyses or machine learning models. By counting the instances of ‘NaN‘ or marked ’unknown’ entries, we can understand potential data deficiencies or gaps that might need addressing by cleaning or imputation methods.

B.7 File Size Distribution

In particular, the exploration of the ‘file_size_kb‘ column was conducted to understand the distribution of file sizes across the dataset. The ‘describe()‘ method provided a summary of these sizes. Understanding file size distribution is significant for evaluating the resource allocation for processing and storage, which might be required at later stages of analysis or deployment.

Throughout this exploration process, insights were recorded into a text file for documentation and future reference. These summaries help form hypotheses that might guide further analyses or data-driven decisions.

B.8 Code Output

data_exploration.txt

```
# Data Size
Number of rows: 34
Number of columns: 5

# Summary Statistics
```

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```
                                                                dataset
                                                                    interleaving_n

                                                                    n_examples

                                                                    file_size_kb


                                                                    filename
count                                                                 34
                34         34           34
                                                34
unique                                                                31
                 5        NaN          NaN
                                                34
top     data_datamix_alexandrainst_foqa_default_train_messages
           unknown        NaN          NaN
    data_alexandrainst_foqa_default_test.jsonl
freq                                                                   4
                26        NaN          NaN
                                                 1
mean                                                                 NaN
               NaN       1795         1508
                                                NaN
std                                                                  NaN
               NaN       2086         1121
                                                NaN
min                                                                  NaN
               NaN         41            9
                                                NaN
25%                                                                  NaN
               NaN        600        636.2
                                                NaN
50%                                                                  NaN
               NaN        848          928
                                                NaN
75%                                                                  NaN
               NaN       1968         2768
                                                NaN
max                                                                  NaN
               NaN       5454         2940
                                                NaN

# Categorical Variables
Column 'dataset' most common value:
data_datamix_alexandrainst_foqa_default_train_messages occurrences
    : 4
```

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```
Column 'interleaving_n' most common value:
unknown occurrences: 26

Column 'filename' most common value:
data_alexandrainst_foqa_default_test.jsonl occurrences: 1

# Missing Values
Column 'interleaving_n' has 26 missing/unknown values
# File Size Distribution
File Size (in KB) Summary:
count      34
mean     1508
std      1121
min         9
25%     636.2
50%       928
75%      2768
max      2940
Name: file_size_kb, dtype: float64
```

C Data Analysis

C.1 Code

The Data Analysis was carried out using the following custom code:

```
# IMPORT
from my_utils import df_to_latex, df_to_figure
import pandas as pd
import numpy as np
import statsmodels.api as sm
from statsmodels.formula.api import ols
import pickle

# LOAD DATA
data = pd.read_csv('interleaving_summary.csv')

# DATASET PREPARATIONS
# Handling missing values by replacing 'unknown' with NaN
data['interleaving_n_clean'] = data['interleaving_n'].replace('
    ↪ unknown', np.nan)

# DESCRIPTIVE STATISTICS
# A descriptive statistics table is needed to understand
    ↪ variability in dataset
## Table df_desc_stat:
caption = "Descriptive Statistics: Number of Examples and File
    ↪ Sizes"
```

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```
df_desc_stat = pd.DataFrame({
    'mean': data[['n_examples', 'file_size_kb']].mean(),
    'std': data[['n_examples', 'file_size_kb']].std()
})
df_to_latex(df_desc_stat, 'df_desc_stat', caption=caption)

# PREPROCESSING
# Creating dummy variables for 'interleaving_n_clean'
data = pd.get_dummies(data, columns=['interleaving_n_clean'],
    ↪ dummy_na=True, drop_first=True)

# ANALYSIS
## Table df_regression_analysis:
caption = "Regression Analysis of File Size on Number of Examples
    ↪ and Interleaving"
formula = "file_size_kb ~ n_examples + interleaving_n_clean_nan"
model = ols(formula, data=data).fit()
df_regression_results = pd.DataFrame({
    'coef': model.params,
    'std_err': model.bse,
    'P>|t|': model.pvalues,
    'conf_low': model.conf_int()[0],
    'conf_high': model.conf_int()[1],
})
df_to_latex(df_regression_results, 'df_regression_analysis',
    ↪ caption=caption)

## Figure df_file_size_variance:
caption = "File Size Variance by Interleaving Status"
grouped_data = data.groupby(data['interleaving_n_clean_nan']).agg(
    n=('file_size_kb', 'size'),
    mean=('file_size_kb', 'mean'),
    std=('file_size_kb', 'std')
).reset_index()
grouped_data['ci_low'] = grouped_data['mean'] - 1.96 *
    ↪ grouped_data['std'] / np.sqrt(grouped_data['n'])
grouped_data['ci_high'] = grouped_data['mean'] + 1.96 *
    ↪ grouped_data['std'] / np.sqrt(grouped_data['n'])
df_to_figure(grouped_data, 'df_file_size_variance', caption=
    ↪ caption, kind='bar', x='interleaving_n_clean_nan',
             y=['mean'], y_ci=[('ci_low', 'ci_high')])

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': data.shape[0],
}
with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

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C.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    """


def df_to_figure(
        df, filename: str, caption: str,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.

     `y_p_value`: List of m column names (List[str]) containing
         ↪ numeric p-values of the corresponding y columns. These
         ↪ numeric values will be automatically converted by
         ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
         ↪ plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.
```

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```
Example:
Suppose, we have:

df_lin_reg_longevity = pd.DataFrame({
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
        ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
    y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
"""
```

C.3 Code Description

The data analysis code undertakes a sequence of steps to explore, preprocess, and analyze the dataset from the file `interleaving_summary.csv`, focusing on the relationship between file size and the number of examples, as well as the possible effect of interleaving status on the file size. Each step is designed to facilitate this exploration and statistical analysis.

C.4 Data Preparation

In the initial step, data preparation is conducted. The column `interleaving_n`, which contains categorical values indicative of interleaving status, includes entries labeled as 'unknown'. These entries are transformed to `NaN` using `numpy`'s `NaN` to conveniently handle missing data in subsequent analyses.

C.5 Descriptive Statistics

To comprehend the variability inherent in the dataset, descriptive statistics are generated for key numerical columns, specifically `n_examples` and `file_size_kb`. This step calculates the mean and standard deviation for these columns, providing preliminary insights into central tendency and dispersion. The results are tabulated using a utility function `df_to_latex`, which formats the DataFrame into a LaTeX table for presentational purposes.

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C.6 Preprocessing

To facilitate the analysis of relationships concerning the interleaving status, the code generates dummy variables for the processed `interleaving_n_clean` column using `pandas`'s `get_dummies` function. This transformation encodes categorical data into a form suitable for regression analysis, ensuring that the effect of missing or unknown interleaving statuses can be statistically examined.

C.7 Statistical Analysis

The core of the analysis is conducted through regression modeling, using `statsmodels`'s ordinary least squares (OLS) regression. A regression model is specified to examine the influence of the number of examples (`n_examples`) and interleaving status (specifically when it is NaN) on file size (`file_size_kb`). The model's parameters, including coefficients, standard errors, p-values, and confidence intervals, are extracted and formatted into a LaTeX table using the `df_to_latex` function, enabling interpretation of the results.

C.8 Visualization

To visualize variance in file size based on interleaving status, a grouped analysis is conducted. The dataset is grouped by the interleaving status indicator, and summary statistics such as count, mean, standard deviation, and confidence intervals are calculated for file size within each group. The results are visualized as a bar chart, created by the `df_to_figure` function, which incorporates error bars denoting confidence intervals, thus illustrating the variance within each group.

C.9 Additional Results Saving

Finally, additional results, specifically the total number of observations, are preserved by serializing them to a file using Python's `pickle` module. This ensures reproducibility and future reference for key characteristics of the dataset derived during this exploratory analysis.

This structured approach ensures a comprehensive understanding of the dataset and systematically probes into testable hypotheses concerning file size, number of examples, and interleaving status.

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C.10 Code Output

df_desc_stat.pkl

```
             mean  std
n_examples   1795 2086
file_size_kb 1508 1121
```

df_file_size_variance.pkl

```
  interleaving_n_clean_nan  n mean  std ci_low ci_high
0                    False  8 2669  350   2426    2911
1                     True 26 1151 1030  754.8    1547
```

df_regression_analysis.pkl

```
                                   coef std_err     P>|t| conf_low
                                        conf_high
Intercept                          1570   300.3  1.11e-05    957.9
          2183
interleaving_n_clean_nan[T.True] -879.7   287.1   0.00449    -1465
        -294.2
n_examples                        0.3399 0.05924  2.59e-06   0.2191
        0.4607
```

additional_results.pkl

```
{
    'Total number of observations': 34,
}
```

D LaTeX Table Design

D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```
# IMPORT
import pandas as pd
from my_utils import df_to_latex, df_to_figure, is_str_in_df,
    ↪ split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES AND FIGURES
all_mapping: AbbrToNameDef = {
    'n_examples': ('Examples', 'The number of examples in each
        ↪ dataset.'),
    'file_size_kb': ('Size (KB)', 'Size of the file in kilobytes.'
        ↪ ),
```

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```
    'interleaving_n_clean_nan': ('Interleaving Info Missing', '
        ↪ Indicates whether interleaving information is missing
        ↪ (1: Yes, 0: No).'),
    'coef': ('Coef.', 'Regression coefficient for each variable.')
        ↪ ,
    'std_err': ('Std. Err.', 'Standard deviation of the regression
        ↪  coefficient estimate.'),
    'P>|t|': ('P-value', 'Probability that the observed
        ↪ relationship is due to chance.'),
    'conf_low': ('95% CI Lower', 'Lower bound of the 95%
        ↪ confidence interval for the coefficient.'),
    'conf_high': ('95% CI Upper', 'Upper bound of the 95%
        ↪ confidence interval for the coefficient.'),
    'ci_low': ('95% CI Lower', 'Lower bound of the 95% confidence
        ↪ interval for the mean.'),
    'ci_high': ('95% CI Upper', 'Upper bound of the 95% confidence
        ↪  interval for the mean.'),
    'n': ('Sample Size', 'Number of datasets considered in each
        ↪ category.'),
}

# Process df_desc_stat:
df_desc_stat = pd.read_pickle('df_desc_stat.pkl')

# Format values:
# Not Applicable

# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_desc_stat, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_desc_stat.rename(columns=abbrs_to_names, index=abbrs_to_names,
    ↪ inplace=True)

df_to_latex(
    df_desc_stat, 'df_desc_stat_formatted',
    caption="Descriptive Statistics for Number of Examples and
        ↪ File Sizes",
    glossary=glossary
)

# Process df_file_size_variance:
df_file_size_variance = pd.read_pickle('df_file_size_variance.pkl'
    ↪ )

# Format values:
# Not Applicable

# Rename rows and columns:
```

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```
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_file_size_variance, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_file_size_variance.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

df_to_figure(
    df_file_size_variance, 'df_file_size_variance_formatted',
    caption="Mean File Size by Missing Interleaving Information",
    note="The figure illustrates the effect of missing
        ↪ interleaving data on the average file size.",
    glossary=glossary,
    kind='bar',
    x='Interleaving Info Missing',
    y=['mean'],
    y_ci=[('95% CI Lower', '95% CI Upper')]
)

# Process df_regression_analysis:
df_regression_analysis = pd.read_pickle('df_regression_analysis.
    ↪ pkl')

# Format values:
# Not Applicable

# Remove 'Intercept'
df_regression_analysis.drop(index='Intercept', inplace=True)

# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_regression_analysis, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_regression_analysis.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

# Shorten long index labels and transpose the table
df_regression_analysis.rename(index={
    'interleaving_n_clean_nan[T.True]': 'Interleaving Missing',
    'n_examples': 'Examples'
}, inplace=True)

# Transpose to make the table narrower
df_regression_analysis = df_regression_analysis.T

df_to_latex(
    df_regression_analysis, 'df_regression_analysis_formatted',
    caption="Regression of File Size on Variables",
    glossary=glossary
)
```

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D.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
        note: str = None,
        glossary: Dict[Any, str] = None,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    `note`: Note to be added below the table caption.
    `glossary`: Glossary for the table.
    """


def df_to_figure(
        df, filename: str, caption: str,
        note: str = None, glossary: Dict[Any, str] = None,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
        xlabel: str = None, ylabel: str = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.
    `note`: Note to be added below the figure caption.
    `glossary`: Glossary for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.
    `xlabel`: Label for the x-axis.
    `ylabel`: Label for the y-axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.
```

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```
        `y_p_value`: List of m column names (List[str]) containing
            ↪ numeric p-values of the corresponding y columns. These
            ↪ numeric values will be automatically converted by
            ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
            ↪ plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.

    Example:
    Suppose, we have:

    df_lin_reg_longevity = pd.DataFrame({
        'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
            ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
        'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
            ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
            ↪ [0.0001, ...],
    }, index=['var1', ...])

    then:
    df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
        ↪ caption='Coefficients of ...', kind='bar',
        y=['adjusted_coef', 'unadjusted_coef'],
        y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
        y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
    """


def is_str_in_df(df: pd.DataFrame, s: str):
    return any(s in level for level in getattr(df.index, 'levels',
        ↪  [df.index]) + getattr(df.columns, 'levels', [df.columns
        ↪ ]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef):
    abbrs_to_names = {abbr: name for abbr, (name, definition) in
        ↪ abbrs_to_names_and_definitions.items() if name is not
        ↪ None}
    names_to_definitions = {name or abbr: definition for abbr, (
        ↪ name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

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D.3 Code Output

df_desc_stat_formatted.pkl

```
\begin{table}[h]
\caption{Descriptive Statistics for Number of Examples and File
    Sizes}
\label{table:df-desc-stat-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lrr}
\toprule
 & mean & std \\
\midrule
\textbf{Examples} & 1795 & 2086 \\
\textbf{Size (KB)} & 1508 & 1121 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Examples}: The number of examples in each dataset.
\item \textbf{Size (KB)}: Size of the file in kilobytes.
\end{tablenotes}
\end{threeparttable}
\end{table}
```

df_file_size_variance_formatted.pkl

```
\begin{figure}[htbp]
\centering
\includegraphics{df_file_size_variance_formatted.png}
\caption{Mean File Size by Missing Interleaving Information
The figure illustrates the effect of missing interleaving data on
    the average file size.
Interleaving Info Missing: Indicates whether interleaving
    information is missing (1: Yes, 0: No).
95\% CI Lower: Lower bound of the 95\% confidence interval for the
     mean.
95\% CI Upper: Upper bound of the 95\% confidence interval for the
     mean.
Sample Size: Number of datasets considered in each category.}
\label{figure:df-file-size-variance-formatted}
\end{figure}
% This latex figure presents "df_file_size_variance_formatted.png
    ",
% which was created from the df:
%
```

Created by data-to-paper (AI)

```
% index,"Interleaving Info Missing","Sample Size","mean","std
    ","95% CI Lower","95% CI Upper"
% 0,0,8,2669,350,2426,2911
% 1,1,26,1151,1030,754.8,1547
%
% To create the figure, this df was plotted with the command:
%
% df.plot(kind='bar', x='Interleaving Info Missing', y=['mean'])
%
% Confidence intervals for y-values were then plotted based on
    column: [('95% CI Lower', '95% CI Upper')].
```

df_regression_analysis_formatted.pkl

```
\begin{table}[h]
\caption{Regression of File Size on Variables}
\label{table:df-regression-analysis-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lll}
\toprule
 & Interleaving Missing & Examples \\
\midrule
\textbf{Coef.} & -879.7 & 0.3399 \\
\textbf{Std. Err.} & 287.1 & 0.05924 \\
\textbf{P-value} & 0.00449 & 2.59e-06 \\
\textbf{95\% CI Lower} & -1465 & 0.2191 \\
\textbf{95\% CI Upper} & -294.2 & 0.4607 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Examples}: The number of examples in each dataset.
\item \textbf{Coef.}: Regression coefficient for each variable.
\item \textbf{Std. Err.}: Standard deviation of the regression
    coefficient estimate.
\item \textbf{P-value}: Probability that the observed relationship
     is due to chance.
\item \textbf{95\% CI Lower}: Lower bound of the 95\% confidence
    interval for the coefficient.
\item \textbf{95\% CI Upper}: Upper bound of the 95\% confidence
    interval for the coefficient.
\end{tablenotes}
\end{threeparttable}
\end{table}
```